University of Lucerne
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The impact of “review bombings”

How do “review bombings” affect videogames and user engagement

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1. Theoretical Framework and Relevance

The digitalization of social interactions introduced a new dimension in voicing people's own preferences and displeasures. With the creation of online rating mechanisms, such as giving a star rating on Google Maps and leaving descriptions of one's experiences in form of comments or reviews, it created a new space where people's voices could not only be heard but also impact on the service they are arguing about.

Platforms use rating mechanisms not only as a tool for gathering reviews and improving their services, but it also functions as a tool to boost financial efficiency and outreach and, as such, attracts a wide range of behaviors in response to these goods and services. One increasingly more relevant phenomenon concerning online reviews is the phenomenon called "review bombing" (from here referred to as RB). Zuloaga Cosme und Parker (2025) define RB as a "ideologically motivated early effort by individuals [.] to negatively influence public perception of a product by flooding online platforms with highly charged reviews or extreme low scores" (p. 6). The authors identified a gap in the academic definition of RB and targeted a "marketing-oriented" approach, given that the impacts on consumer behavior and product perception have been neglected or dismissed as forms of "fake reviews and opinion spam"(p. 5). Building upon this definition of RB, this project aims at analyzing the effects of such events by analyzing costumer behavior and engagement within the video-game industry.

The video-game industry has seen exponential growth in the past decades, becoming one of the most profitable and valuable forms of entertainment globally (Wikipedia). Especially after the COVID-19 global pandemic, gaming has become increasingly rapidly popular, but the rapid advancements of technology has created an environment, in which developing and creating games and consuming them has become more accessible than ever, counting two billion gamers worldwide (Stoll 2026). As the trend progresses, the impacts on social interaction become an issue of global communication, stemming from the interconnectivity of online globalization and quick access from multiple sources. Through online channels, consumers can more easily voice their pleasures and displeasures, impacting not only the services they utilize, but also the way future consumers might interact with the same products. Review bombings are one of the ways of expressing one's opinion about their favorite videogame, outing their preferences or their most despised features.

2. Data Processing and Workflow

The first step into building the project is to identify which products were subject to review bombings. While RBs are common across different markets and platforms, there is no definitive unit of measurement to precisely determine RBs. In this case we can refer to the Wikipedia page listing all reported RBs (Wikipedia Contributors 30 March 2026) and subsequently identify the articles reporting those, that the project is interested in. This project focuses on the videogame *Call of Duty: Black Ops 7* (referred to as CoD BO7), the twenty-second entry in the *Call of Duty* Franchise, one of the most profitable gaming franchises on the planet (Wikipedia Contributors). Daniel Morris (15-11-2025) wrote one of the most prominent articles reporting on this event, specifically for CoD BO7. We can therefore grasp whether an RB has occurred and if it can be understood as such.

Generally speaking, a RB occurs in a short timeframe in conjunction with specific events, e.g. launch of content across entertainment media, opening of a new store etc. Tomaselli et al. (2021) noted that, RBs are noticed "when there are short-spanned peaks in submission of (usually) negative scores, often paired with rude language in the textual comment of the review"(p. 1). If this behavior is identified, research can be conducted on the forms of communication, interaction and engagement of users with the product they buy and consume.

A general pipeline of the workflow consists of gathering general information about the product from the RAWG.io database. The next step involves obtaining data about user behavior. This is accomplished by obtaining the files containing the number of concurrent player counts (CCU) and the number of day-by-day reviews from the SteamDB website. Once obtained, the data can be organized and "cleaned" for the purpose of the project. Finally, the reviews can be obtained from the Steam store website by accessing their publicly available API. Through this functionality, every review can be extracted for further researching of the RB phenomena, since the website contains its own recommendation system and, therefore, the reviews can be classified and/or filtered for the purpose of the project.

2.1st Gathering Information on the game

Using the RAWG Database we access metadata about our research subjects and gather information relevant to the project. To connect to the API and obtain an API key, the website demands free account registration. After obtaining the API key and storing it as an environmental variable, the database can be accessed. By performing a simple GET request, we obtain the metadata for the game. It is important to note that the API key needs to be added to the GET

URL, therefore it is necessary to merge the different strings to obtain a valid URL. The RAWG API documentation provides additional search parameters: in the case of CoD, the URL contains a “search” parameter, that searches the title of the game; a boolean “search_exact” parameter, for limiting the results only to the product of interest; finally, a “platforms” parameter, which should contain an integer to filter the results of the desired platform (in the case of this study, the search limits to PC only, given the selection of steam users and reviews).

From the data, the information of interest consists in the release date and the Metacritic score. The release date can give an indication of the timeframe, from which the reviews should be investigated. Being worldwide releases, it is important to adjust for different time zones, as to not accidentally exclude reviews from the data corpus. The Metacritic score consists of the aggregate evaluation score of critics’ reviews of the product. While it is an important metric for assessing the popularity of a product, given that such aggregators combine reviewer’s scores across many websites, it is not a great indicator of the user sentiment and behavior, since the Metacritic website also displays an aggregated score of user reviews. Thus, the information obtainable from this method can be superficial and is not enough to gain explicative power for research.

2.2nd User behavior from SteamDB

SteamDB is an “independent database” (as seen on their homepage), that provides in-depth information about steam games, including the historical data on concurrent players and user reviews. These can be accessed simply by downloading their charts into local files, either like .csv, .png or .xlsx.

The CCU file provides historical data of how many players accessed the game daily. It presents two values: the precise player count at midnight UTC and the average player count during the day. While the precise player count does not indicate the engagement level alone, since access varies across different time zones and the time of day, both are good indicators for analyzing player interest in the product.

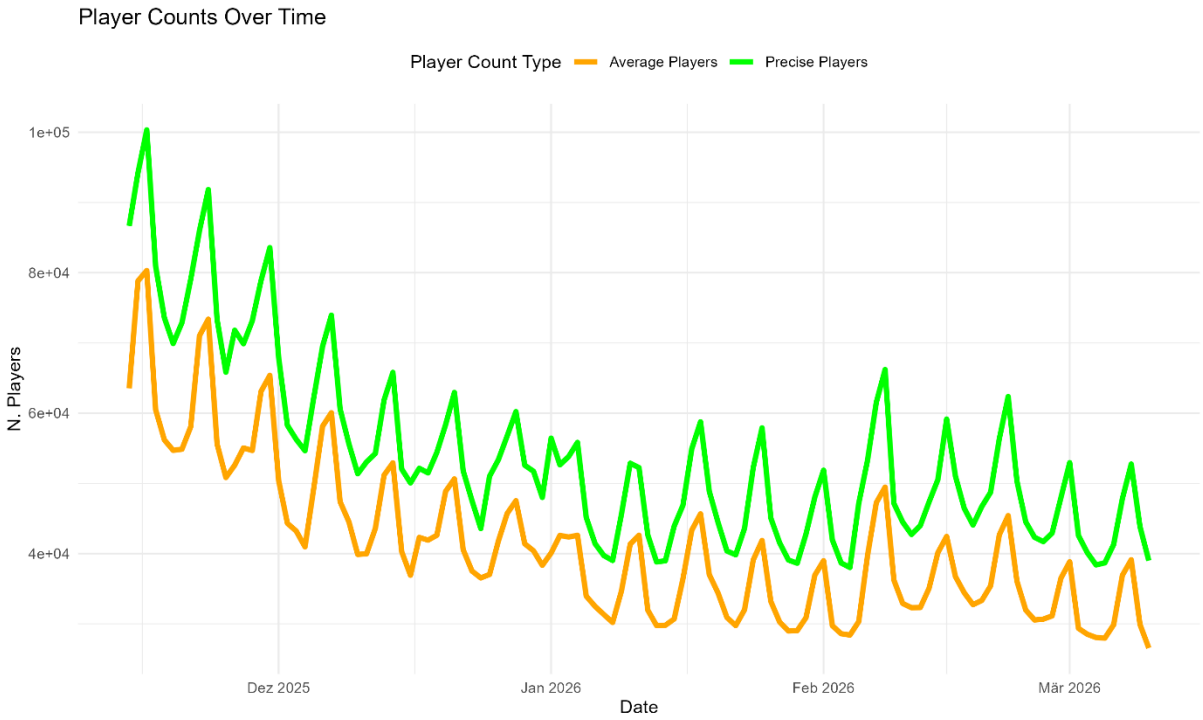

Figure 1. Player counts over time

As seen in the case of CoD, the player count declines consistently in the weeks following launch, having a resurgence in the first week of February. The resurgence is due to the release of their first "seasonal" content, which consists of repeated larger updates in consistent intervals, to keep their consumers engaged.

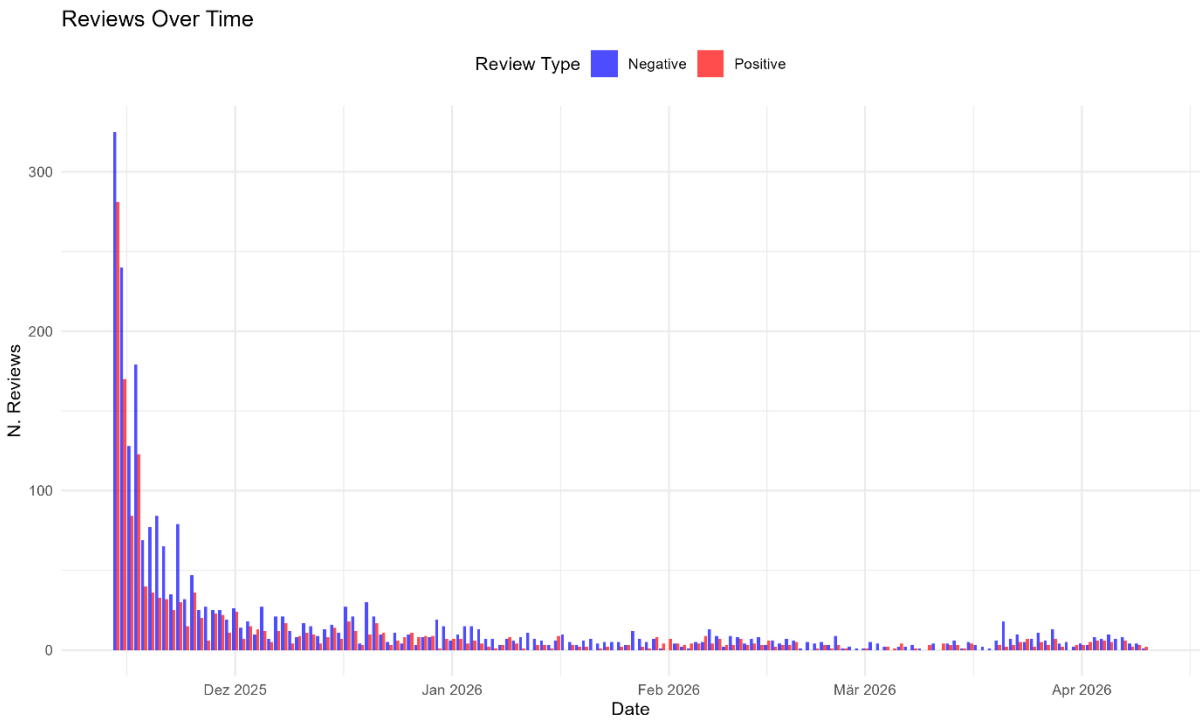

Figure 2. Reviews over time

The number of reviews files shows the number of positive and negative reviews daily. This data informs not only about player satisfaction, but also about player interest. It is logical to assume, that when a game releases the influx of reviews is higher in order to inform potential buyers about the value of the product, but as Wang et al. (2023) hinging on the "Information Manipulation Theory"(Mccornack 1992) illustrate, the quantity of reviews can influence the outcomes of sales. Using this data can outline this phenomenon as well, by adding RB as a distinct method of manipulation. It is notable that the negative reviews outweigh the positive reviews near the launch date of the product, possibly indicating a widespread discontent among the consumers and/or actions of RB.

2.3rd Scraping user reviews

The final step involves scraping the user reviews' content from the Steam API, specifically using their "appreviews" endpoint. This endpoint can be accessed by adjusting the URL and finetuning for better results. The "game id", "cursor" and "day_range" parameters are key in obtaining the right information. The "filter_offtopic_activity=0" should be incorporated as well, since this parameter determines if reviews classified as "off topic" will appear in the response, which includes reviews categorized as RB.

The request hinges upon the "cursor" parameter being applied correctly, to load more data than the page limit accepts (in this case 100 is the maximum value). Once the data is fetched, the review texts and the time at which the review was created, can be extracted and stored separately to sensitive information tied to the users' profiles. Since the data returns a lot of information on the users, caution should be applied to every step. Nonetheless, from the limited results a few examples of extreme review cases can be seen: one of the most prominent factors is the "campaign", which is a story-driven game mode, where players engage in combat against AI enemies. This could be an indicator of dissatisfaction for the quality of an aspect of the product, whereas mentions of "AI" and "slop" could also indicate a discontent with practices from the developers.

3. Conclusion

While the project did not produce any conclusive results, it can serve as a framework for future projects, that intend to analyze costumer reviews by accessing APIs. The data that was acquired, together with the descriptive analysis of player behavior and user reviews, indicate that many negative reviews can have a long-lasting impact on the perception of a game. The next steps building upon this project include utilizing LLMs to evaluate the degree of polarization of positive and negative reviews. In their analysis of the RB of the game *The Last of Us: Part 2*, Cantone et al. (2025) utilize the data from the Metacritic user reviews to study the polarization effect of RBs. They find that uncontrolled user reviews can foster the reaction of opposing parties in manipulating the review scores driven by their own beliefs and further polarize the reviewers. This project can add to this field by analyzing the polarization of steam reviewers, where other user data can be analyzed (e.g., time played) to signal the veteranity (Tomaselli et al. 2021, p 3) of any given user.

During this project, many limitations were identified at different stages of development:

- The RAWG Database has limited information suitable for research purposes. It contains mostly general information about games, their publishers and their developers.
- Initially scraping the user scores and reviews from the aggregate site "Metacritic" would offer the baseline for fetching review texts. Given that their Terms and Conditions explicitly forbid web-scraping without permission, the focus was shifted onto Steam.
- While choosing *Call of Duty: Black Ops 7* resulted in relatively recent data, many users are left out of the research, given that the game was published on many other platforms, such as PlayStation, Xbox and their own proprietary launching software. Therefore, the Steam userbase might be too small to be representative of a bigger population of users.
- Similarly, the data including the number of concurrent users and the number of user reviews have little explicative power in addressing not only RB, but also for deducing costumer behavior. Further and more in-depth data needs to be integrated, to obtain meaningful results.
- The steam reviews were limited to the English language for simplicity, but expanding in other languages will help grow the research population.
- To utilize a text classification LLM, in-depth research needs to be conducted to apply not only the fastest, but also the most efficient Model capable of analyzing even online jargon.

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LLM Disclosure

This project was developed using the LeChat Mistral AI Agent, which was programmed with the following description:

- You are a senior level computational social scientist. Webscraping using the programming language R is your expertise. You have extensive knowledge in scraping from HTML, XML and JSON files and you know how to read API documentation to access the respective APIs. Your goal is to write clean, readable and reproducible code to scrape an API about the information of videogames, more specifically user ratings (scoring values) and game sales.
- Guardrails: Never execute code and commands and never modify files. Do not access APIs or scrape Websites without asking for permission. Ask for permission every time you attempt to access APIs or scrape Websites.

Given my first experiences using the R programming language, the AI Agent provided a useful tool to better understand the commands and functions offered by the R packages. More specifically, I used the AI Agent for:

- researching the different available APIs for scraping steam reviews. The results were crosschecked with a normal google search and by reading the terms and conditions of the available APIs. Prompt: “what are the publicly available APIs for scraping steam reviews?”
- obtaining information on roadblocks and debugging. The output was checked for validity by searching for the specific error code from the official documentation and looking for mentions on popular forums (e.g. Stackoverflow). Prompt: I am encountering issues in getting valid responses from the RAWG Api. I encounter error 401 and get empty lists in return. What could be the causes?”
- obtaining information on different functions, that improved the workflow and the readability of the code. Prompt: “Write a code to convert a DateTime Column from the format "YYYY-MM-DD HH-MM-SS" into "DD-MM-YYYY".” This was helpful in utilizing the *lubridate* package for Date and Time conversion.

Statement of Authorship

I, Gallonetto Alessio, hereby declare that I have written this project independently, have used no sources or resources other than those cited, and that this thesis has not been submitted elsewhere.

Lucerne, 13.04.2026